

S&P 500 Drawdown Prediction: Feature-Engineering Catalog for Two

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TL;DR

- The core thesis is well-supported: corrections and bear market distinct events with largely distinct macro drivers (recession/credit/earnings/labor force participation) and 6-12 months slow macro variables (yield curve, credit spreads, excess corrections (10-20%) are driven by fast positioning/sentiment informative at weeks-to-3-months and hairfdeerrfndatfuorreesass; expect higher AUC on the bear model.
- BEAR priorities (rough evidence for) forward spread (Engstrom 2019), 10y-3m term spread (Estrella-Mishkin probit 1998), bond premium / HY OAS, Sahm-rule unemployment momentum, 6-month growth leads and 6-months of unemployment rate (ALFRED).
- CORRECTION priorities term-structures slope (VIX3M-VIX) and z-score deterioration (%>200DMA, SPX vs 200DMA, 12-1 momentum), financing (Chicago Fed NFCI/ANFCI), valuation extremes (CAPM), sentiment (AAII, put/call) as contrarian extreme dummies, Hindenburg Omen (weak OOS) and LPPL. Corrections are common — design for modest AUC and class imbalance.

This report is a literature-grounded feature catalog; it documents observables, sources/series IDs, functional forms with rationale (evidence strength, citations). It does not fit models. Coefficient = (-) / - on a trailing window; “percentile” = trailing empirical; “6m/YoY change” = differences / log-differences; all transforms are standardized.

Key Findings: The Predictability Landscape and the

Bear markets are easier to predict than ordinary returns, and strategies in macro/recession were shown. “Predicting the Bear Market: Macroeconomic Variables as a Leading Indicator of Bear Market” 33(2):211-223 (Markov-switching bear definition, monthly SP yield-curve term spread and the effects to predictors; the term spread accurate beyond six months, short-rate/inflation within three months, the user’s 12m vs 6m problem. Henri Nyberg, “Predicting Bear Markets with Dynamic Binary Time Series Models”, (’9):3351-3363, confirms that adding an autoregressive/lagged-state term materially improves performance. & Chou, “Further Evidence on Bear Market Predictability: The Bond Premium”, International Review of Economics & Finance 10(6):1121-1141. The

- Note uses EBP for 12m recession prob. Isolates credit-supply Strong.
- High-yield FRED: BAMLH0A0HYM2 Form: level + 3m change + z-score (positive/strong).
 - BAA-AAA default FRED: BAAAd.AA(A or BAA10Y). Form: level / 6m change positive. Best at 1-3m for bears (Chen-Chen-Chou 2017). (BAA-Y10) — note collinearity (below).
- Labor momentum (strong, coincident-to-slightly-leading)
 - Sahm rule FRED: SAHMRREALTIME real-time vintage LEAD for 3m UUA of U3 minus its trailing-12m moving average (don't binarize) + opt [≥ 0.5m]ny. Sign: positive. Signals onset ~3m in. Sahm (201) for strong.
 - Initial claims FRED: DUCSuw.LC4W\$A Form: YoY % / deviation from 1 Sign: positive - strong.
 - Composite leading indicators (strong).
 - Conference Board LEI 6-Trend LEI (RwhitUSISyLEADs a related proxy of FRED). Form: annualized 6m growth dummy [6m growth < -1% AND diffusion ≤ 50]. Sign: negative. LEI leads turns ~7 months. Note 2022 — declines, no NBER recession) — use context clues, not a hard t.
 - ISM Manufacturing ISM (Primary source). Form: distance from 50 orders — investment spread is cleaner. Sign: negative. Manuf — one input among many - strong.
 - Policy stance & earnings (moderate).
 - Real fed funds / FRED: Ff/FEDFUNDSnus trailing inflation / Tan Sign: positive; 6m month change in Fed FFR (sharp cuts = warning) Tokic-Ja Moscor)ate.
 - Earnings: EPS growth, margins S&P 500 earnings FRED: Nhl.PA profits CP). Form: YoY growth; revisions diff of interest Sign: negative.
 - Building permits FRED: PERMIT, YoY %. Sign: Moderate.
 - M2 growth FRED: M2SL, YoY real. Sign: negative weak contra moderate/mixed (unstable post-2020).
 - Term premium FRED: 10y (NY Fed). Sign: ambiguous (Bernanke-Chyr) ate/mixed.

Catalog 2 — CORRECTION Model (10-20% drawdown, 6-mo)

Mechanism: positioning, sentiment, valuation stresses, pleiq-u centiles, spreads, thresholds idlummgi windows; informative hor months (6m is the long end).

- Volatility term structure & vol-of-vol (strong/moderate)
 - VIX term-structure FRED: VIX3M(-VIXr's feature ratio VIX/VIX3M, z-score (1-2yr). Sign: negative on spread tion (ratio > 1) = stress. ConM ad hroa-t 80% of d n g e .
 - VIX level / FRED: VIX Low VIX (complacency) w across re c thi ognhs ; VIX is coincident — model nonlinear Model by percentile bins).
 - VVIX FRED: VIX level / z-score, Reagni p e d + t m p e i e d R V o 2 l 1 s p r l e a d sign posM d d e a t e .
- Trend & breadth (moderate — robust regime conditioner).

- SPX vs 200DMA % deviation / 200-1yr [< 200] dummy. Sign: negative. Moskowitz, Ooi & Pedersen, JFE 2005, 104, 10, 228-250 (1-12m persistence). Recommend lengthening user's MA. Moderate.
 - 12-1 month trailing 12m return ex last month. Sign: negative.
 - Breadth % > 20 SDMA ChasStPSXA 2/0 SETH. Form: perdcievntrigleen ee (breadth falling while index rising). Sign: negative. Use active. Moderate.
 - A/D line, McClellan's ENHANCED NYMOS NYHS Sign: negative. Heinden-burg Overak / mixed OOS — exclude; as tpsrd in a very intense turbulent more robustly by a breadth-divergence term.
3. Financial conditions & liquidity (moderate-strong).
- Chicago Fed NFC FRANKFCCI, ANR (weekly from 1971; pre-z-score SD 1). Form: level + 4 - ANR (width of gap, normalized to economy) is correction signal. Sign: positive (NtFiCgHrRtleSrK-, hN fQ bGR E DSIUb, - NFC I LEVERANC Moderate-strong).
 - Funding / money-market interest rate spread (EDRAT (archive), SOFR-OIS, spreads. Level / z-score of esriagtrne p(ossipatrivee)..
4. Valuation (moderate conditioner; weak standalone timer).
- Shiller SCAPELer Yale dataset / mpuelrtcpelntciolme./ E s more (long w NOT raw level (drift). Sign: positive. Weak short-term timer feature—convert to percentile, use Wesako (ntdi mti inogn) e/r m o d e a a (severity).
 - Equity risk premium E/P - 10y real (TIPS). Sign: negative. (2008) OOS with sign Mroedsetraitcet.ions.
 - Buffett indicator (mkt/CAP/ GDP, 2008), Tobin's q, cointile / detrend sign positive moderate.
5. Sentiment & positioning (moderate—contrarian extremes,
- AAI bull AAI weekly. Extreme decile dummy (contrarian, nonlinear AAI's own data: extreme bullishness predicts a 6m decline positive at extreme. Webauk-Imioschenreastse./ mixed.
 - Put/call (CBOE CPO / CPEz)-score/extreme dummies; low put/call ratio → higher future drawdown. Pan & Poteshman, "The Info for Future Stock Returns", F2006 Moderate.
 - NAAIM, Investors Intelligence scores US Midealing moderate.
 - Margin debt ChianRA monthly debit balance growth / deviation from trend (NOT level—non-stationary). Sign: positive. Few prices (largely weak mode case).
 - CFTC COT, dealer gamma, CBOE SKEW, spec z-score; SKEW level Mixed/weak CBOE states SKEW is "not designed to predict crash minor percentile feature.

Shared vs. Distinct Predictors — and Why

Distinct to BTEFASR, 10y-3m/10y-2y, EBP, BAA-AAA, Sahm/claims, PMI, real fed funds/easing, earnings contraction, permits, M 6-18m. Chen (2009) / Nyberg (2013) / Tokic-Jacobsaornm(a2r0k23t) show correction → bear transition is a credit-risk + policy-easing

Distinct to CORREX ItON m structure, VVIX, RV-IV, breadth oscillations, ANFCL/funding spreads, AAI/put/call/NA

dealer gamma. Mean-reverting, days-to-weeks; produced 10-20% progress without a recession/credit crunch.

Legitimately shared (transformed differently):

- Credit spreads/earnings (credit spread) - z-score - rapid widening (liquidity shock). Toxic-Jackson's level deepens drawdowns past 20% is the bridge.
- CAPE/valuation (ibeyar) vulnerability to correction; percent both; weak standalone timing.
- Trend (200 DMA) and Volatility from a regime term; correction windows.

Design one credit-spread variable appears in both (different segregated by speed. This also reduces multicollinearity and distinction.

Mathematical Definitions of Engineered Features

Let SPX_{close} ; running $max_{k \leq t}$ Peak $drawdown = \frac{SPX_{close} - \max_{k \leq t} SPX_{close}}{\max_{k \leq t} SPX_{close}}$ /Peak 1(user's). Forward max drawdown over horizon

$$= \min_{k \leq t} \left(\frac{SPX_{close} - \max_{k \leq t} SPX_{close}}{\max_{k \leq t} SPX_{close}} - 1 \right).$$

- CORRECTION target: $1\{-0.20 < \text{drawdown} < -0.10\}$.
- BEAR target: $1\{\text{drawdown} < -0.20\}$ (earliest: BEAR = any $> 20\%$; CORRECTION = max-drawdown landing in 10-20% with no $> 20\%$ breach.)

BEAR features:

$$\begin{aligned} &= 10 - 3, \quad 1\{\text{drawdown} < 0\} \\ &= (6 - 3) \\ \Delta_3 &= \frac{SPX_{close} - \max_{k \leq t} SPX_{close}}{\max_{k \leq t} SPX_{close}} - 3 \\ &= \frac{SPX_{close} - \max_{k \leq t} SPX_{close}}{\max_{k \leq t} SPX_{close}}, \quad = \frac{SPX_{close} - 60}{60} \\ &= \frac{SPX_{close} - 3}{3} - \min_{-11 \leq t \leq 11} \frac{SPX_{close} - 3}{3} \\ &= \left(\frac{SPX_{close} - 6}{-6} \right) - 1, \quad 1\{\text{drawdown} < -0.04 \text{ if } SPX_{close} \leq 50\} \\ &PMI - 50, \quad (\text{new orders inventory}) \end{aligned}$$

CORRECTION features:

$$\begin{aligned} &= \frac{SPX_{close} - 3}{3}, \quad = \frac{SPX_{close} - 3}{3}, \quad 1\{\text{drawdown} > 1\} \\ &= \frac{SPX_{close} - 1}{200}, \quad 1\{\text{drawdown} < 200\}, \end{aligned}$$

$$\Delta_{12} = \frac{\Delta_1 - \Delta_2}{\Delta_3} \times 100\%$$

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CAPE (user's, explicit):

$$\Delta_{12} = \frac{\Delta_1 - \Delta_2}{\Delta_3} \times 100\%$$

— Campbell & Shiller, "Stock Prices, Earnings, and Excess Returns", *Journal of Finance*, 1988, 43(3): 661–676.

Feature-Engineering Best Practices for These Two B

1. Stationarity. Transform every trending series (price, raw CAPE, levels) to spreads, changes/growth, MA deviations, or trailing and diffusion indices are naturally stationary and preferred.
2. Real-time vs revised (e.g., FRED vs BEA). Use revised macro (LEI, U3, ISM, permits, GDP, payrolls) to avoid look-ahead (e.g., Jan later 5.4%). SAHMRREALTIME > revised SAHM. Market data until publication timing (FINRA margin debt lags weeks).
3. Lag selection. Use an explicit "as-of" calendar. For example, if using 12-month rolling average, use 11-month lag. Trailing windows only (no overlapping).
4. Standardization. Expand window for long rolling (5–10 yr) trailing using only past data — never full-sample (leaks future information). Standardized on a fixed sample; re-standardize on a trailing sample.
5. Overlapping observations. Moving averages of 12 months outcomes (share 5/6 or 11/12 of the window), in fact, in regression, use 11 months (Newey & West, 1994, 55(3): 703–708) or 12 months (SPESS, 1992, 35(3): 357–386) for overlapping long-horizon data. Richardson & Whitelaw, "The Myth of Long-Horizon Data", *Journal of Finance*, 1995, 50(2): 1605 (cross-horizon estimators ~99% correlated under the null hypothesis is not independent). For logistic: also try non-overlapping, block-bootstrap inference, and a lagged-statistic.
6. Class imbalance / rare events. Data is highly imbalanced. King & Zeng, "Logistic Regression in Rare Events Data", *Political Analysis*, 2001, 9(2): 137–163. Use rare events probabilitites; apply their finite-sample bias correction or case-control sampling. Also, use a logit score, accuracy, calibrate (Platt / isotonic) on held-out sample and pick the best model (avoid test-set Youden look-ahead).
7. Multicollinearity. Create a family (BAA–AAA, HY OAS, EBP) in the EBP orthogonalizes default risk, so EBP + one market spread is a good family (10y–3m, 10y–5y, 10y–15y, 10y–20y, 10y–25y, 10y–30y, 10y–35y, 10y–40y, 10y–45y, 10y–50y, 10y–55y, 10y–60y, 10y–65y, 10y–70y, 10y–75y, 10y–80y, 10y–85y, 10y–90y, 10y–95y, 10y–100y). NTFS + at most one conventional measure move together — one divergence term. NFCL subindices correlate with the parent

features, check VIF, prefer one well-transformed representation over many, regularization given the short effective sample (few independent observations).

Additional Predictors Beyond the User's Original List

Beyond the user's eight (yield curve, credit spread, Sahm, breadth, CAPE, drawdown):

Add to BEAR: NTFS (highest-priority; dominates 10y-3m EBP Engstrom (Gilchrist-Zakrajsek 2012; Chen-Henrichsen 2017); unrevised; Chen-Chen (2017); new-orders-inv; initial claims momentum (CSA-Yearly) (lagged funds) (Tokic-Jackson).

Add to CORRECTION: VIX/VIX3M ratio + backward-difference dummy (implies volatility from 200DMA + 12-1 month moving average; breadth-divergence (indicator replacement for the Homeny); ANFCI + 4-week change; forward E/P - ratio (Thompson 2008); put/call z-score and AAI extreme points (Man 2006; three not linearly independent YoY (growth) — endogenous condition/interaction).

De-prioritized/excluded (weak/mixed OOS): high-frequency use breadth divergence (seven-parameter fit, fails own race Fantazzini-Geraskinchuk-Skewness (saturation); a lot of information not designed to predict crashes — minor predictive power (2017) (supply/demand) (post-2020).

Optional advanced (if data available): S&P positioning (z-score), gamma exposure (proprietary — strong intuition, thin public), exponentiality proxy (second difference of trailing log price substitute).

Recommendations

1. Stage 1 — build the BEAR model first signal, higher expected AUC. 5 features: NTFS (level), HY OAS (level + 3m change), EBP (12-1 month growth). Use ALFRED vintages, Newey-West/Hodrick-Sargent event correlation; test <AUC 0.705 are likely mis-lagging; standardization; the recession-probit literature.
2. Stage 2 — build the CORRECT model. VIX term-structure slope > 200DMA + divergence, SPX % dev from 200DMA, ANFCI, CAPE peak and AAI extreme points. Modest AUC (~0.55-0.65) which implies these signals are weak out-of-sample; do not over-tune. If sample but collapses out-of-sample, that is the overfitting.
3. Shared-variable discipline: exactly one credit-spread variable in different transforms (level/6m-change for BEAR; short z-score for other families segregated by speed).
4. Validation: forward/expanding-window OOS only; report PFT plot; choose thresholds on training data. Re-examine windows if OOS degrades sharply.
5. Thresholds that would change the design: red > 5 standard deviations EBP + HY OAS only; if the bear model's lagged-state term absorbs, prefer a dynamic-probit-style specification (Nyberg 2013); coefficients flip sign across folds, drop them.

Caveats

- Evidence asymmetry: BEAR / recession-prediction literature reviewed (Estrella–Mishkin, Engstrom–Sharpe, Gilchrist CORRECTION / technical literature is thinner and more practical signals (Hindenburg Omen, SKEW, dealwre g & nonram in xærdg in out-of-sample nudpæ or fagged accordingly).
- Few independent tests have been only a handful of post-1960 and recessions, so effective sample size is small regardless of how to regularize and resist feature proliferation.
- Regime instability: several relationships are documented to be unstable; structural breaks; the 2022–23 LEI false positive; M2 post-2008 down that was preceded by an inverted 10y–3m or NTFS — the motivation (the Tokic–Jackson line of work). Use continuous features or regime shifts.
- Endogeneity & publication bias: it and broader topics rather than lead them; macro series are revised. Both bias naïve ALFRED / real-time and publication-lag discipline above is.
- I was unable to verify a couple of secondary citations to the ISM/LEI series availability on FRED for the official TCB LEI Board data rather than the Philly Fed USSLIND proxy) — construction. Where I report institutional/practitioner figures (e.g., thresholds), treat them as descriptive heuristics, not predictions.